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PROFESSIONAL SUMMARY

I am a hands-on Security Engineer and Backend Developer with nearly 2+ years of professional experience spanning cybersecurity, full-stack backend development, agentic AI systems, and technical leadership. I have sole-authored and deployed production-grade backend systems, designed end-to-end security architectures, and led cross-functional development teams, all within a fast-paced startup environment where I operated across engineering, product, and client-facing roles simultaneously. My technical depth covers offensive security, cloud security on AWS and Azure, DevSecOps, E2EE systems, vulnerability assessment, and AI-driven security automation. I actively build custom tooling and agentic workflows that improve detection, response, and operational efficiency, leveraging LLMs and automation as force multipliers rather than replacements for engineering judgment. Beyond engineering, I bring research credibility with 6 published papers across cybersecurity and ML, 100,000+ cumulative blog reads, and a track record of translating complex security concepts into practical, scalable systems. I approach every problem with an architecture-first mindset, a bias toward building, and the ability to own outcomes end to end, from design to deployment to client delivery.

SKILLS

Hard Skills:

- **Programming & Scripting:** Python (automation, security tooling, agent workflows), Bash/Shell scripting, Complete E2EE API Backend, basic JavaScript for web testing and tooling.
- **Red Teaming & Offensive Security:** Web application penetration testing, system exploitation, privilege escalation, vulnerability assessment, attack surface analysis, proof-of-concept development.
- **Network VAPT:** Network enumeration, service fingerprinting, vulnerability analysis, lateral movement analysis, network penetration testing.
- **Cloud Security & DevSecOps:** AWS and Azure security fundamentals, Infrastructure as Code (Terraform), container security (Docker, Docker Compose), CI/CD security concepts, misconfiguration analysis, cloud threat modeling, cloud cost optimization, Cloudflare security and DNS management.
- **Linux & Systems:** Linux system administration, package management, process and service management, shell scripting, security hardening, debugging, and performance analysis.
- **AI & Agentic Systems:** Agentic AI workflow design and implementation, modular control pipelines, autonomous security detection and response systems, integration with LLMs (OpenAI GPT, Meta Llama, Ollama), agent-based automation, threat analysis augmentation, intelligent parsing and classification systems.
- **E2EE & Secure Systems:** End-to-end encrypted API backend development, secure communication architecture, privacy-focused system design.
- **Security Tooling:** SBOM and supply-chain security tools (Trivy, Syft), vulnerability scanners, container scanning tools, static analysis concepts, and custom tool development.
- **Research & Technical Writing:** Writing research papers, conference papers, and technical documentation; long-form security blogs with 100,000+ cumulative reads; LaTeX for academic and professional documentation.
- **Databases & Data Handling:** MongoDB, Supabase, SQLite3, JSON-based data modeling, structured logging, dataset preprocessing for security and ML pipelines.
- **APIs & Backend Systems:** REST API design, authentication concepts, secure API development, full production backend development, integration of security and AI services.
- **Leadership & Management:** Team coordination and mentorship, sprint planning and management, product lifecycle management, client relationship management, architectural decision-making.

Soft Skills:

- Technical Communication
 - Problem Solving
 - Project Ownership
 - Client Management
- Team Collaboration
 - Critical Thinking
 - Adaptability
 - Technical Leadership

EDUCATION

4/2021 - 3/2024	Bachelor of Computer Applications - Jain (Deemed-To-Be-University) Scored CGPA - 8.7	Undergraduate
4/2018 - 3/2021	PUC(SEBA) - Presidency University Scored 87.6%	PUC

WORK EXPERIENCE

10/2024 – Present **Security Engineer & Backend Lead**

Cygne Noir Cyber

- Sole backend developer for a production dating application — independently designed, architected, and deployed the entire backend system as a single contributor. Capable of scaling from thousands to millions of concurrent users.
- Designed and implemented a comprehensive security architecture, including an autonomous threat detection, monitoring, and automated ban system for real-time application protection.
- Built a fully-featured client admin command center for operational management, user oversight, and platform control.
- Owned end-to-end cloud infrastructure — deployment, maintenance, cost optimization, and Cloudflare security configuration across AWS and Azure environments.
- Led and coordinated a 3-person front-end development team, managing task delegation, code reviews, and development velocity throughout the project life-cycle.
- Managed client relationships directly — conducting requirement gathering sessions, handling change requests, and maintaining close communication with stakeholders on deliverables and priorities.
- Implemented Sprint management and product life-cycle planning, driving a faster and more structured development workflow across the team.
- Designed and implemented agentic AI workflows and modular control pipelines to support security automation, threat analysis, and research-driven use cases.
- Conducted end-to-end product testing and application security testing across all verticals.
- Worked across 6+ product research initiatives, converting 4 into MVP projects and 1 into a fully live production deployment.
- Contributed to cross-domain research in E2EE systems and secure agentic AI, expanding technical scope beyond core cybersecurity.
- Represented the firm in university talks and led intern and employee training sessions.
- Performed compliance checks and ensured regulatory and security compliance across all deployed environments, identifying and remediating gaps proactively using many tools and techniques.
- Conducted infrastructure cost simulations and optimization analysis, identifying critical S3 and network cost inefficiencies. Implemented targeted optimizations that achieved a 90% reduction in projected cloud infrastructure costs at a million-user scale.
- Authored detailed backend architecture plans for upcoming development phases, ensuring technical continuity, scalability, and clear implementation road-maps for future team execution.

03/2024 – 10/2024 **Freelance Researcher & Developer**

Independent

- Designed proof-of-concept implementations for security research, automation workflows, and experimental tooling.
- Developed custom automation scripts for testing, monitoring, and data processing use cases.
- Provided technical writing and review support for research papers, reports, and documentation.
- Designed and implemented a complete Point of Sale system for a restaurant venture, along with deployment of the complete backend on a local system, including making networking adjustments and edge system deployment.
- Designed and implemented a design studio for a designer who is commissioning the sale of third-party products on Amazon; this includes public-facing deployments at very low cost.

07/2023 – 03/2024 **Offensive Security Engineer Intern**

Avercyber Technologies

- Performed cloud security assessments across AWS and Azure environments, focusing on misconfigurations and exposure risks.
- Conducted vulnerability research and penetration testing on infrastructure and application components.
- Built and deployed security-focused Infrastructure-as-Code environments using Terraform for testing and research.
- Researched Linux initialization processes, system hardening techniques, and supply-chain security tooling (SBOM).

05/2023 – 07/2023 **Cybersecurity Engineer Intern**

Avercyber Technologies

- Developed Python-based red team and security automation tools to support assessment workflows.
- Assisted with vulnerability assessments, reporting, and remediation validation.
- Conducted research on AWS security rules and early-stage AI applications in cybersecurity.

PROJECTS

GPT-Vuln-Analyzer

[Github Link](#)

- Python
 - AI / LLMs
 - Security Analysis
- Built an AI-assisted proof-of-concept for vulnerability analysis using large language models.
 - Integrated multiple LLM backends including Llama2, GPT, PaLM, and Ollama for comparative analysis.
 - Implemented features such as DNS reconnaissance, subdomain enumeration, and structured vulnerability reasoning.
 - Designed the system to be modular and extensible for integration into larger security pipelines.

Startup-SBOM

[Github Link](#)

- Python
 - Linux
 - SBOM
- Reverse-engineered the Linux boot and initialization process to extract startup behavior of system packages.
 - Analyzed RPM and DPKG package metadata to identify service initialization and startup persistence.
 - Implemented chroot-based inspection to accurately evaluate boot-time behavior without full system execution.
 - Produced a structured SBOM-style output highlighting potential startup and persistence vectors.

QuadraInspect

[Github Link](#)

- Python
 - Mobile Security
 - Reverse Engineering
- Developed an automated Android (APK) security analysis framework for vulnerability detection.
 - Integrated multiple analysis techniques to inspect permissions, components, and application behavior.
 - Focused on scalable analysis and structured reporting for mobile security assessments.
 - Designed the framework to support extensibility for future analysis modules.

Brute Framework

[Github Link](#)

- Python
 - Windows
 - Security Framework
- Recreated the core concepts of PentestBox by reverse-engineering its architecture and rebuilding it from scratch.
 - Developed an all-in-one Windows-based security testing framework focused on portability and ease of use.
 - Implemented centralized tool management with upgrade-friendly and accessible configurations.

Nmap-API

[Github Link](#)

- Python
 - API Development
 - Network Security
- Built a RESTful API wrapper around Nmap as part of a graduation project.
 - Enabled programmatic network scanning and result retrieval via structured endpoints.
 - Focused on usability, extensibility, and integration with external systems.

HackBot

[Github Link](#)

- Python
 - AI / LLMs
 - Assistant Systems
- Developed an LLM-powered assistant for security-related analysis and information gathering.
 - Integrated GPT-based models to assist with code inspection and scan result interpretation.
 - Designed the system to support researchers and ethical hackers during analysis workflows.

CVE-LLM-Dataset

[Github Link](#)

- Dataset
 - LLM Research
 - CVE Analysis
- Created a structured dataset as a proof-of-concept for training and evaluating LLMs on CVE-related data.
 - Focused on dataset design challenges specific to cybersecurity and vulnerability intelligence.

C2C-Server

[Github Link](#)

- Python
 - C2
 - Red Team
- Developed a command-and-control (C2) server to demonstrate real-world attack and communication patterns.
 - Used the project for controlled red-team experimentation and learning scenarios.

• Python • WiFi • Windows Internals	WinFiHack	Github Link
	• Implemented WiFi brute-force experimentation using native Windows networking libraries. • Focused on understanding Windows wireless stack behavior and automation constraints.	
• Android • Containers • Full-Stack	Komo.do-Hub	Github Link
	• Developed a mobile application interface for the Komodo container management platform. • Worked across frontend and backend components using Java, Node.js, React Native, and React. • Focused on usability and operational visibility for containerized environments.	

CERTIFICATIONS

Valid 6-2023 to 6-2027	Certified Ethical Hacker - V12 Successfully completed my CEH V12 Certification.	EC-Council
Valid 6-2023 to 6-2027	Certified Network Defender Successfully completed my CND Certification.	EC-Council

PUBLICATIONS

Cybersecurity & ML	API-Based Network Scanning This paper presents an API-driven approach to network vulnerability scanning, based on the implementation and findings of the Nmap-API project. The work focuses on improving stability, usability, and scalability of network scanning systems through structured resource management and virtualized client interactions.	Paper Link
Cybersecurity & ML	AI-Based Enumeration and Exploit Suggester This research proposes the integration of artificial intelligence into cybersecurity workflows to automate reconnaissance, vulnerability enumeration, and exploit suggestion. The paper explores the feasibility of using AI models to assist in identifying potential attack vectors and prioritizing vulnerabilities.	JETIR
Cybersecurity & ML	AI in Action: Exploiting the Nexus of Cybersecurity This paper examines the role of artificial intelligence in modern cybersecurity environments, with a focus on integrating AI-driven analysis into CI/CD pipelines to enhance security automation, threat detection, and response workflows.	Paper Link
Cybersecurity & ML	Using Autoencoders for Malware Detection This study investigates the application of autoencoder-based machine learning models for malware detection. The work analyzes how unsupervised learning techniques can be used to identify anomalous behavior in executable and network data.	FMDB
Cybersecurity Cloud Computing	&Docker-Based Decentralized Vulnerability Assessment This paper explores a decentralized vulnerability assessment framework built using Docker-based containerization. The approach focuses on distributed scanning, scalability, and the integration of AI-assisted techniques for port scanning and vulnerability analysis.	FMDB
Network Security ML	&ML-Driven Secure Communication for 6G Networks This work presents a machine learning-driven approach to secure communication in next-generation 6G networks. The paper discusses how ML techniques can enhance security, reliability, and adaptability in emerging high-speed network infrastructures.	Springer

HONORS

Competition	Certificate of Merit Won first place in a research presentation competition hosted by BGS College of Engineering and Technology.	BGS
Honor	Certificate of Appreciation Awarded by Jain(Deemed To Be University) for being the speaker in the IRDCSTEM-2023 post-conference learning.	JAIN